SELF SUPERVISED REPRESENTATION LEARNING FOR ORGANOID IMAGES

Masked Siamese Networks

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Introduction

- Organoids are biological structures used to study morphological development
- Large-scale microscopy produces thousands of heterogeneous organoid images
- No clear labels $\rightarrow$ problem is **clustering**, not classification
- Learn image representations using **self-supervised learning**
- Multiple SSL approaches exist $\rightarrow$ Used **Masked Siamese Networks (MSN)**

Example

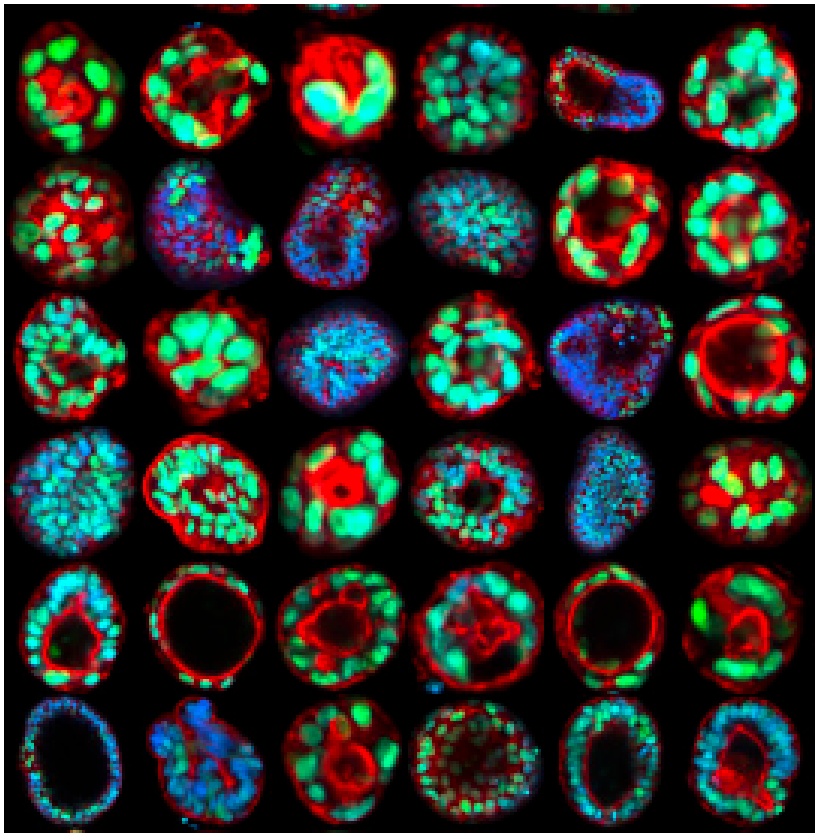


Figure 1: Organoid Images

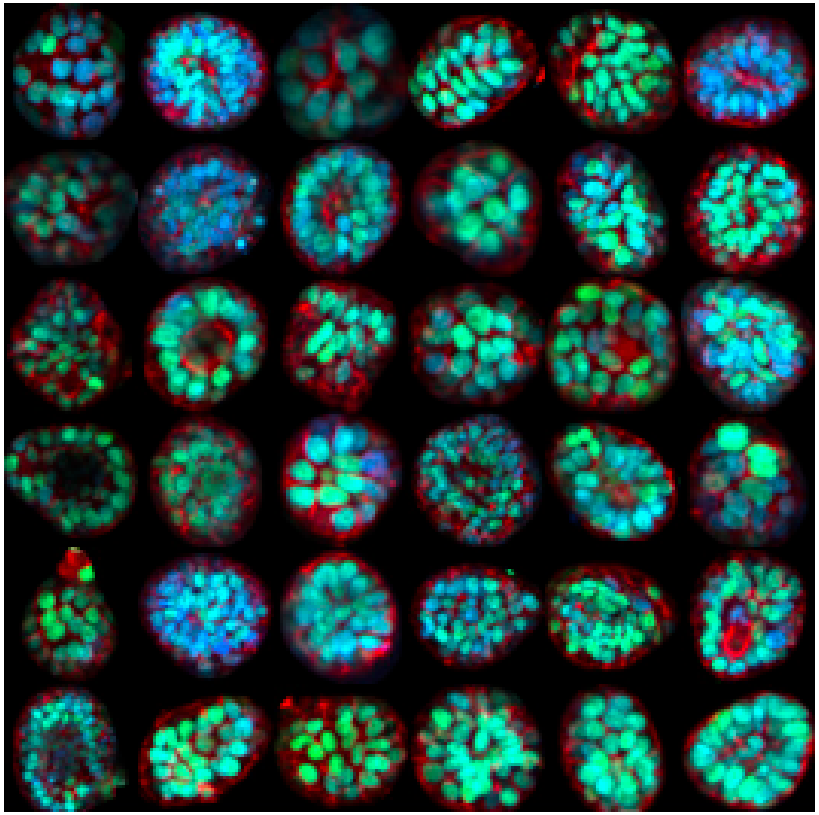


Figure 2: Clustered Organoid Images

Motivation

- Need representations capturing **morphology and structure**
- Contrastive methods either require negative samples or focus on pixel-level reconstruction, not semantic similarity
- MSN learns semantic similarity directly from masked and unmasked views
- Masked-view training encourages **robust, semantic features**
- No pixel reconstruction $\rightarrow$ focus on high-level similarity

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Masked Siamese Networks (MSN): Core Idea

- **Goal:** Learn semantic representations from unlabeled organoid images
- **Siamese setup:** Two views of the same image are created
  - **Target view**: unmasked
  - **Anchor view**: heavily masked
- A shared Vision Transformer encodes both views
- **Training objective:**
  - Match the representation of the masked view to the unmasked view
  - Minimize the loss function

Masked Siamese Networks (MSN): Core Idea

- **Prototype-based predictions:**
  - MSN uses K learnable prototype vectors $q \in \mathbb{R}^{K \times d}$
  - Probability distribution computed through softmax
- **ME-MAX Regularization:**
  - Loss function: $\mathcal{L} = \mathcal{L}_{\text{align}} - \lambda H(\bar{p})$
  - where $\mathcal{L}_{\text{align}} = \frac{1}{MB} \sum_{i=1}^{B} \sum_{m=1}^{M} H(p_i^+, p_{i,m})$
  - and, $\bar{p} = \frac{1}{MB} \sum_{i,m} p_{i,m}$
- **Prevents Prototype collapse**

Masked Siamese Networks (MSN): Architecture

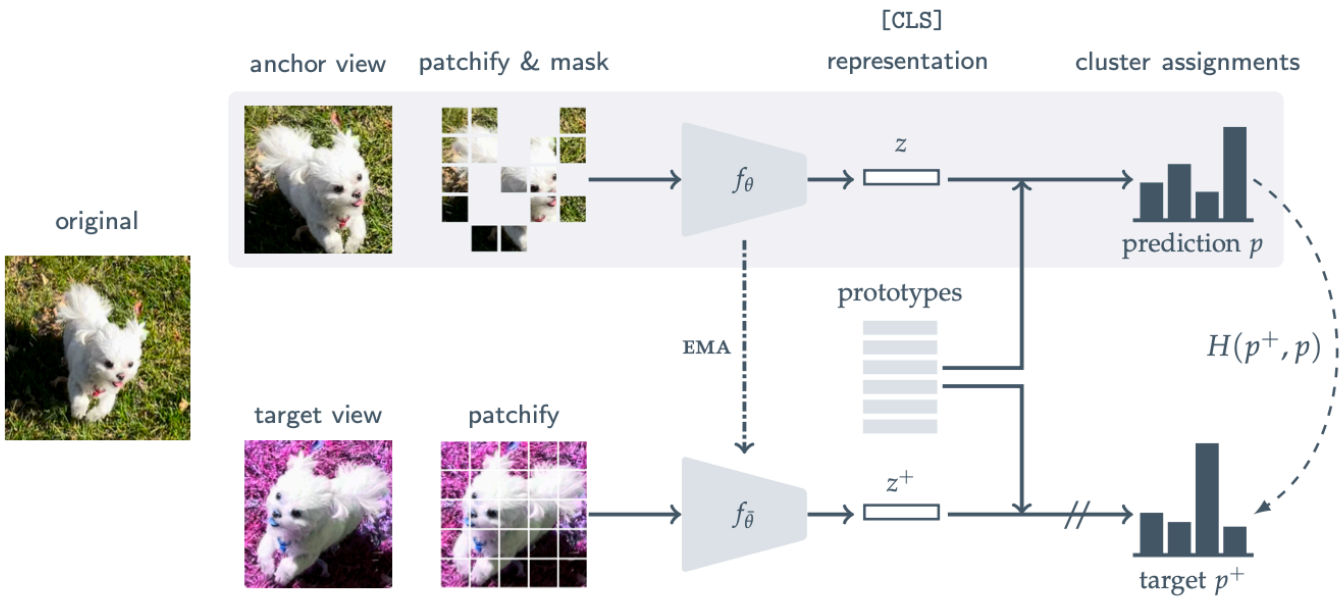


Figure 3: MSN Architecture Example

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Methodology: MSN Training

- **Dataset:** 80,000 unlabeled organoid images for self-supervised training
- **Key adaptation:** Excluded color distortion
- **Augmentations:** Random crops, flips, rotations, Gaussian noise, elastic deformations
- **Image resolution:** 128 x 128 (reduced from 224 x 224 to preserve small organoids)
- **Training setup:** Focal cropping (0.5 to 0.65 scale)

Methodology: Feature Extraction

- **Input preprocessing:** 128 x 128 image input, normalized with current dataset stats
- **Encoder:** Frozen MSN Vision Transformer
- **Embedding extraction:** [CLS] token output from ViT used as image representation
- **L2 normalization:** All embeddings normalized to unit length ($z_{\text{norm}} = z/\|z\|_2$)
- **Output:** Cosine similarity between embeddings in range [-1, 1]

Methodology: Clustering

- **Formulation:** Correlation clustering problem (no predefined cluster count)
- **Cost computation:** $c_{ij} = m - \cos(g(x_i), g(x_j))$
- **Algorithm:** Greedy union-find
  - Sort all pairs by cost
  - Iteratively merge pairs with $c_{ij} < 0$ using union-find
- **Advantage:** Automatically determines cluster count from pairwise similarities

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Results: Quantitative Analysis

- **Test-100 (In-distribution):**
  - ARI: 0.6705, NMI: 0.8065, Accuracy: 0.7900
  - Recovered 10/10 clusters from Test-100
- **Test-30 (Out-of-distribution):**
  - ARI: 0.5123, NMI: 0.6083, Accuracy: 0.7667
  - Recovered 3/3 clusters from Test-30

Results: Qualitative Analysis

- Representative cluster shown from MSN embeddings
- **5 out of 6** organoids belong to same morphological class
- Demonstrates biologically meaningful similarity
- Tolerates natural intra-class variability

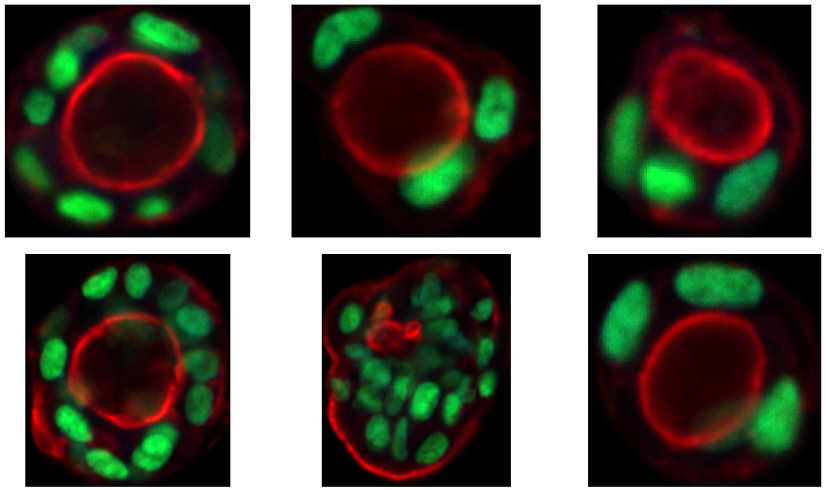


Figure 4: Single cluster example

Results: Qualitative Analysis

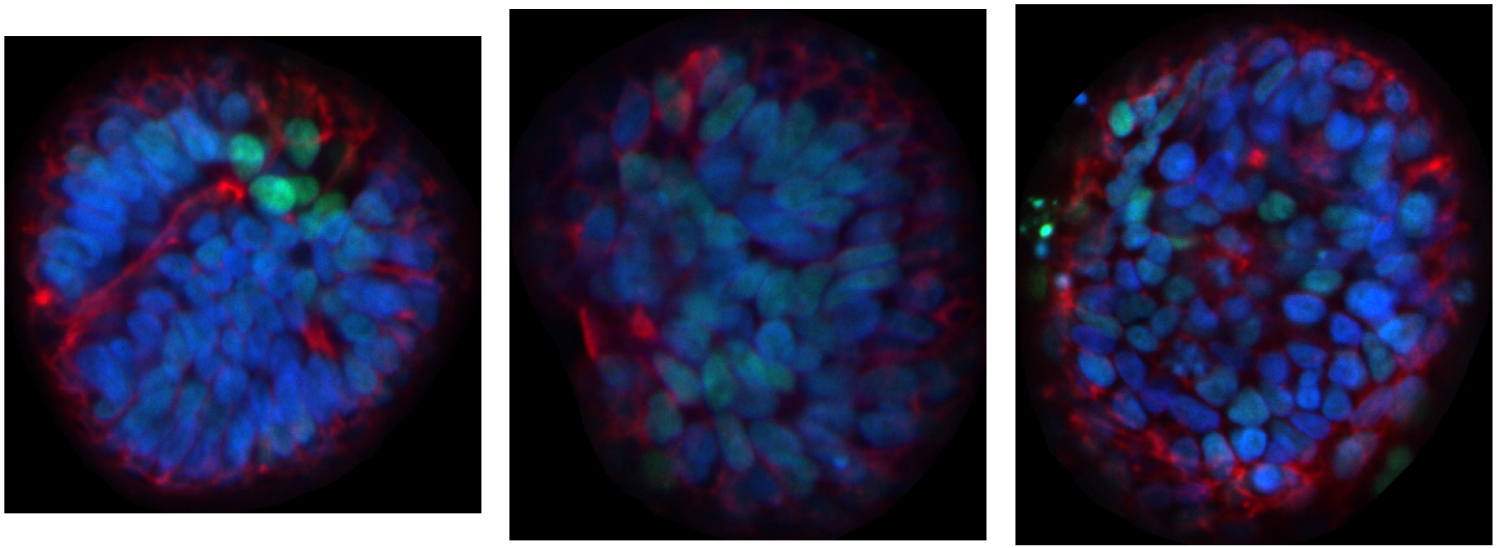


Figure 5: Single cluster examples

Results: Qualitative Analysis

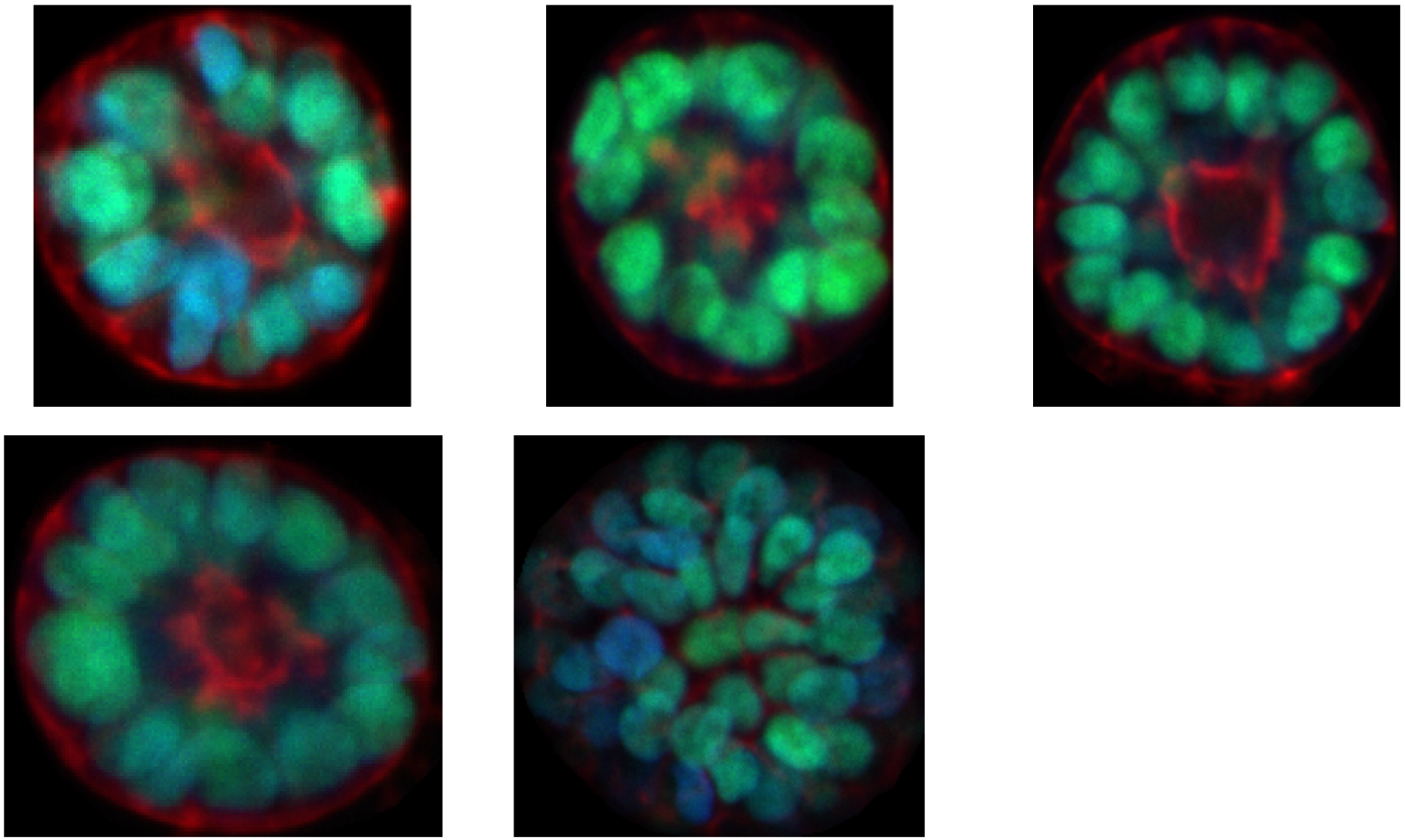


Figure 6: Single cluster examples

Results: Key Findings

- Achieved 79% accuracy on Test-100 and 76.6% accuracy on Test-30
- Correlation clustering correctly inferred cluster numbers
- Critical domain adaptations for capturing morphological features:
  - Preserving color information
  - Optimizing resolution (128 x 128)
  - Choosing masking and focal metrics
  - Ensuring maximum prototype usage
  - Choosing correct λ

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✓Conclusion

- **Achievement:** MSN successfully learns meaningful representations for organoid clustering without labeled training data
- **Performance:** Good results on in-distribution (ACC: 0.79) and reasonable generalization to unseen classes (ACC: 0.7667)
- **Key advantage:** Eliminates expensive manual annotations, scalable to large datasets
- **Limitations:** Excluding color augmentation may limit robustness; needs longer training
- **Future work:** Increase batch size, more prototypes, managed aggresiveness of augementations

Thank You!